

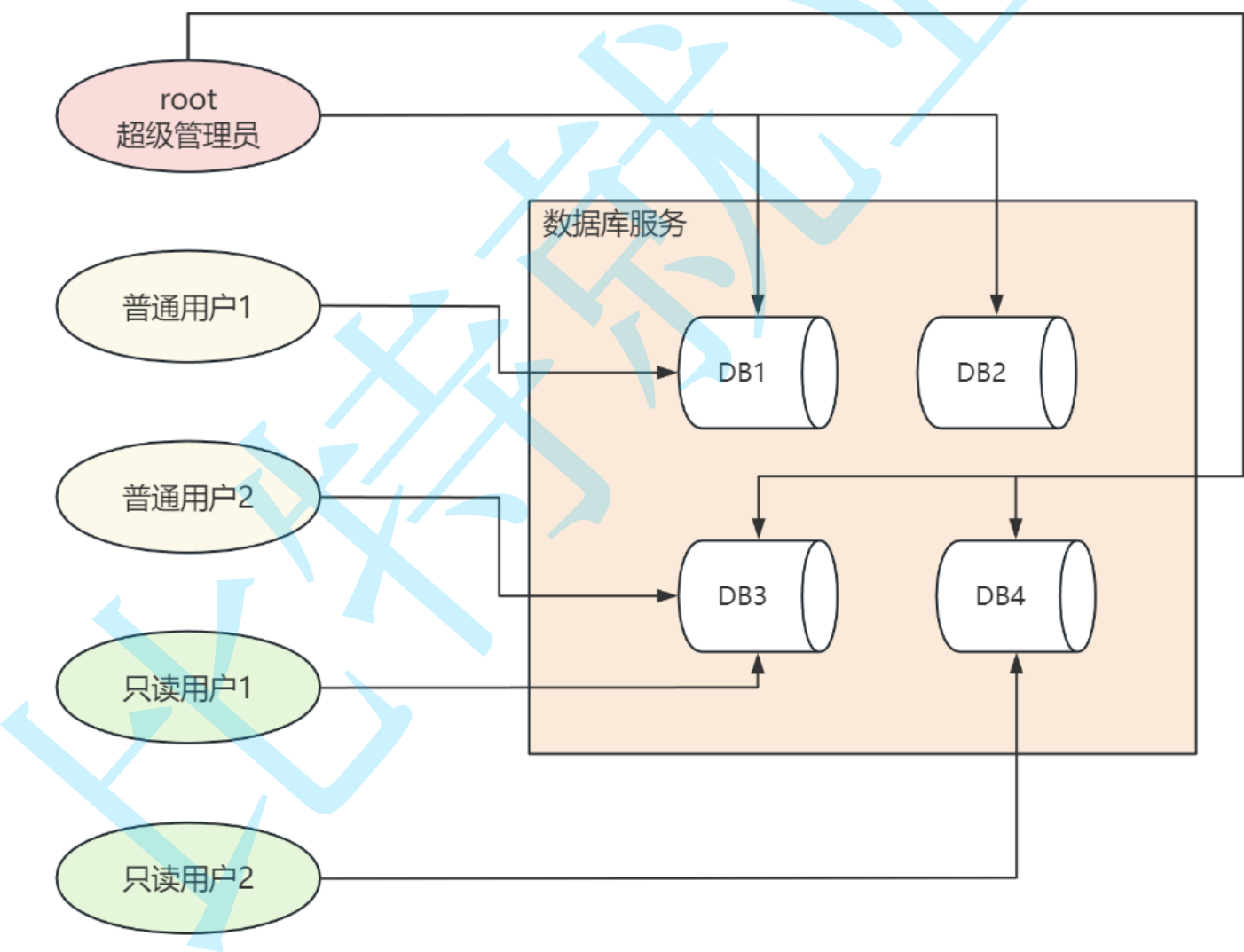
12. 用户和权限管理

1. 本节目标

- 了解数据库的用户与常用操作
- 了解数据库权限与授权

2. 应用场景

数据库服务安装成功后默认有一个root用户，可以新建和操纵数据库服务中管理的所有数据库。在实际的使用过程中，通常每个应用对应着一个数据库，我们只希望某个用户只能操纵和管理当前应用对应的那个数据库，而不能操纵和管理其他应用的数据库，这时就可以添加一个用户并指定用户的权限



如上图所示：

- `root` 可以访问和操纵所有的数据库：DB1, DB2, DB3, DB4

- **普通用户1** 只能访问和操纵数据库DB1
- **普通用户2** 只能访问和操纵数据库DB3
- **只读用户1** 只能访问数据库DB3
- **只读用户2** 只能访问数据库DB4

3. 用户

3.1 查看用户

MySQL的用户信息保存在mysql系统数据库的user表中，可以通过Select语句查看，如下所示：

```

1 # 选择数据库
2 mysql> use mysql
3 Database changed
4
5 # 查看所有的表
6 mysql> show tables;
7
8 # 查看表结构
9 mysql> desc user;
10
11 #查询user表中的记录
12 mysql> select host, user, authentication_string from user;
13 +-----+-----+-----+
14 | host          | user          | authentication_string |
15 +-----+-----+-----+
16 | localhost    | mysql.infoschema | $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED |
17 | localhost    | mysql.session    | $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED |
18 | localhost    | mysql.sys        | $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED |
19 | localhost    | root           | $A$005$4"/zI3
    ,S2!905H8kQgTq2Boe6PFB0p6cbLIqLX0TasCS58WzBBdXrC |
20 +-----+-----+-----+
21 4 rows in set (0.00 sec)

```

- **host**: 允许登录的主机，相当于白名单，如果是localhost，表示只能从本机登陆

- **user**: 用户名
- ***_priv**: 用户拥有的权限
- **authentication_string**: 加密后的用户密码

3.2 创建用户

3.2.1 语法

```
1 CREATE USER [IF NOT EXISTS] 'user_name'@'host_name' IDENTIFIED BY  
  'auth_string';
```

user_name: 用户名，用单引号包裹，区分大小写

host_name: 主机或IP(段)，用单引号包裹

auth_string: 真实密码，有些密码策略不允许使用简单密码

3.2.2 注意事项

- 如果不指定host_name相当于'user_name'@'%', %表示所有主机都可以连接到数据库，**强烈建议不要这样设置**，因为会导致严重的安全问题
- user_name和host_name分别用单引号包裹，如果写成'user_name@host_name'，相当于'user_name@host_name'@'%'
- host_name可以通过子网掩码设置主机范围
 - 198.0.0.0/255.0.0.0: A段网络中的任意一台主机
 - 198.51.0.0/255.255.0.0: 198.51 B段网络中的任意一台主机
 - 198.51.100.0/255.255.255.0: 198.51.100 C段网络中的任意一台主机
 - 198.51.100.1: 只包含特定IP地址的主机
- 从MySQL 8.0.23开始，指定为IPv4地址的主机值可以使用CIDR表示法写入，例如198.51.100.44/24
- 允许在IP地址中使用%通配符，比如，主机值'%匹配任何主机名，198.51.100.% 匹配198.51.100 C段网络中的任何主机。MySQL 8.0.35中已弃用，以后可能会删除

3.2.3 示例

```
1 # 添加一个名为bit的新用户，允许从本机登录  
2 mysql> create user 'bit'@'localhost' identified by '123456';  
3 Query OK, 0 rows affected (0.02 sec)  
4
```

```
5 # 添加一个名为bit1的新用户，允许从192.168.1.1/24网段登录
6 mysql> create user 'bit1'@'192.168.1.1/24' identified by '123456';
7 Query OK, 0 rows affected (0.02 sec)
8
9 # 查询，添加成功
10 mysql> select host, user, authentication_string from user\G
11 ***** 1. row *****
12             host: 192.168.1.36/24
13             user: bit1
14 LYkbuBpcAPQ1f2JJSc8Je4u.5ce6UPmQ01U2pHmfzKQ7-:;%
15 ***** 2. row *****
16             host: localhost
17             user: bit
18 authentication_string:
19 $A$005$P+EPe6Ht`5S%XoXIjzgBT5EuT2nYza1lHaP6f0aSw2B2he7EaFnFzB
20 ***** 3. row *****
21             host: localhost
22             user: mysql.infoschema
23 authentication_string:
24 $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED
25 ***** 4. row *****
26             host: localhost
27             user: mysql.session
28 authentication_string:
29 $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED
30 ***** 5. row *****
31             host: localhost
32             user: mysql.sys
33 authentication_string:
34 $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED
35 ***** 6. row *****
36             host: localhost
37             user: root
38 authentication_string:
39 $A$005$4"/zI3S2!905H8kQgTq2Boe6PFB0p6cbLIqLX0TasCS58WzBBdXrC
40 6 rows in set (0.00 sec)
```

```
1 # 使用新用户登录
2 C:\Users\bit>mysql -ubit -p
3 Enter password: *****
4 Welcome to the MySQL monitor.  Commands end with ; or \g.
5 Your MySQL connection id is 16
6 Server version: 8.0.39 MySQL Community Server - GPL
7
8 Copyright (c) 2000, 2024, Oracle and/or its affiliates.
```

```
9
10 Oracle is a registered trademark of Oracle Corporation and/or its
11 affiliates. Other names may be trademarks of their respective
12 owners.
13
14 Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

3.2.4

3.4 修改密码

3.4.1 语法

```
1 # 为指定用户设置密码 【推荐】
2 ALTER USER 'user_name'@'host_name' IDENTIFIED BY 'auth_string';
3
4 # 为指定用户设置密码
5 SET PASSWORD FOR 'user_name'@'host_name' = 'auth_string';
6
7 # 为当前登录用户设置密码
8 SET PASSWORD = 'auth_string';
```

3.4.2 示例

```
1 # 以root身份登录，为'bit'@'localhost'用户重置密码
2 mysql> ALTER USER 'bit'@'localhost' IDENTIFIED BY '987654';
3 Query OK, 0 rows affected (0.02 sec)
4
5 # 以root身份登录，为'bit'@'localhost'用户重置密码
6 mysql> SET PASSWORD FOR 'bit'@'localhost' = '123456';
7 Query OK, 0 rows affected (0.01 sec)
8
9 # 以bit用户登录，并确认当前登录用户，USER()与CURRENT_USER()是同义词
10 mysql> select USER();
11 +-----+
12 | USER() |
13 +-----+
14 | bit@localhost |
15 +-----+
16 1 row in set (0.00 sec)
17
18 # 修改当前登录用户的密码
```

```
19 mysql> SET PASSWORD = '111111';
20 Query OK, 0 rows affected (0.02 sec)
21
```

3.5 删除用户

3.5.1 语法

```
1 DROP USER [IF EXISTS] 'user_name'@'host_name'[, ...];
```

3.5.2 示例

```
1 # 删除用户'bit1'@'192.168.1.1/24'
2 mysql> drop user 'bit1'@'192.168.1.1/24';
3 Query OK, 0 rows affected (0.01 sec)
4
5 # 删除成功
6 mysql> select host, user, authentication_string from user;
7 +-----+-----+-----+
8 | host      | user      | authentication_string
9 +-----+-----+-----+
10 | localhost | bit1      | $A$005$P+EPe6Ht`5S%KxoXIjzgBT5EuT2nYza1lHaP6f0aSw2B2he7EaFnFzB |
11 | localhost | mysql.infoschema | $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED |
12 | localhost | mysql.session   | $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED |
13 | localhost | mysql.sys       | $A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED |
14 | localhost | root         | $A$005$4"/zI3
    ,S2!905H8kQgTq2Boe6PFB0p6cbLIqLX0TasCS58WzBBdXrC |
15 +-----+-----+-----+
16 5 rows in set (0.00 sec)
```

4. 权限与授权

- MySQL内置支持的权限列表4.1



Privilege	Grant Table Column	Context
ALL [PRIVILEGES]	Synonym for “all privileges”	Server administration
ALTER	Alter_priv	Tables
ALTER ROUTINE	Alter_routine_priv	Stored routines
CREATE	Create_priv	Databases, tables, or indexes
CREATE ROLE	Create_role_priv	Server administration
CREATE ROUTINE	Create_routine_priv	Stored routines
CREATE TABLESPACE	Create_tablespace_priv	Server administration
CREATE TEMPORARY TABLES	Create_tmp_table_priv	Tables
CREATE USER	Create_user_priv	Server administration
CREATE VIEW	Create_view_priv	Views
DELETE	Delete_priv	Tables
DROP	Drop_priv	Databases, tables, or views
DROP ROLE	Drop_role_priv	Server administration
EVENT	Event_priv	Databases
EXECUTE	Execute_priv	Stored routines
FILE	File_priv	File access on server host
GRANT OPTION	Grant_priv	Databases, tables, or stored routines
INDEX	Index_priv	Tables
INSERT	Insert_priv	Tables or columns
LOCK TABLES	Lock_tables_priv	Databases
PROCESS	Process_priv	Server administration
PROXY	See proxies_priv table	Server administration
REFERENCES	References_priv	Databases or tables
RELOAD	Reload_priv	Server administration
REPLICATION CLIENT	Repl_client_priv	Server administration
REPLICATION SLAVE	Repl_slave_priv	Server administration
SELECT	Select_priv	Tables or columns
SHOW DATABASES	Show_db_priv	Server administration
SHOW VIEW	Show_view_priv	Views
SHUTDOWN	Shutdown_priv	Server administration
SUPER	Super_priv	Server administration
TRIGGER	Trigger_priv	Tables
UPDATE	Update_priv	Tables or columns
USAGE	Synonym for “no privileges”	Server administration

4.1 给用户授权

刚创建的用户没有任何权限，需要给他授权，如下所示：

```
1 mysql> select user();
2 +-----+
3 | user() |
4 +-----+
5 | bit@localhost |
6 +-----+
7 1 row in set (0.00 sec)
8
9 # 看不到其他的数据库
10 mysql> show databases;
11 +-----+
12 | Database |
13 +-----+
14 | information_schema |
15 | performance_schema |
16 +-----+
17 2 rows in set (0.00 sec)
```

4.1.1 语法

```
1 grant priv_type[, priv_type ...] on priv_level
2 to 'user_name'@'host_name' [WITH GRANT OPTION]
```

- **priv_type**: 根据类型，参考根据列表4.1中的Privilege列
- **priv_level**: * | *.* | db_name.* | db_name.tbl_name | tbl_name，比如*.*表示所有数据库下的所有表
- **'user_name'@'host_name'**: 指定用户
- **[WITH GRANT OPTION]**: 可选，允许用户将自己的权限授权给其它用户

4.1.2 示例

- 为bit@localhost用户授权于 java01 数据库的 select 权限

```
1 # 授权
2 mysql> grant select on java01.* to 'bit'@'localhost';
3 Query OK, 0 rows affected (0.04 sec)
4
```

5 # 查看数据库，可以看到java01

6 mysql> show databases;

```
7 +-----+
8 | Database          |
9 +-----+
10 | information_schema |
11 | java01             |
12 | performance_schema |
13 +-----+
14 3 rows in set (0.00 sec)
```

15

16 # 选择数据库

17 mysql> use java01

18 Database changed

19

20 # 查看所有表

21 mysql> show tables;

```
22 +-----+
23 | Tables_in_java01   |
24 +-----+
25 | account            |
26 | class              |
27 | course             |
28 | emp                |
29 | exam               |
30 | for_delete         |
31 | score              |
32 | student            |
33 | student1           |
34 | t_check            |
35 | t_recored          |
36 | t_recored_old      |
37 | t_truncate         |
38 | users              |
39 | v_student_socre    |
40 | v_student_socre_v1 |
41 | v_student_total_points |
42 +-----+
43 17 rows in set (0.00 sec)
```

44

45 # 查询数据

46 mysql> select * from student;

```
47 +-----+-----+-----+-----+-----+-----+-----+
48 | id | name   | sno   | age | gender | enroll_date | class_id |
49 +-----+-----+-----+-----+-----+-----+-----+
50 | 1  | 唐三藏 | 100001 | 18  | 1      | 1986-09-01  | 1        |
51 | 2  | 孙悟空 | 100002 | 18  | 1      | 1986-09-01  | 1        |
```

```

52 | 3 | 猪悟能 | 100003 | 18 | 1 | 1986-09-01 | 1 |
53 | 4 | 沙悟净 | 100004 | 18 | 1 | 1986-09-01 | 1 |
54 | 5 | 宋江 | 200001 | 18 | 1 | 2000-09-01 | 2 |
55 | 6 | 武松 | 200002 | 18 | 1 | 2000-09-01 | 2 |
56 | 7 | 李逵 | 200003 | 18 | 1 | 2000-09-01 | 2 |
57 | 8 | 不想毕业 | 200004 | 18 | 1 | 2000-09-01 | 2 |
58 +-----+
59 8 rows in set (0.02 sec)
60
61 # 写入一条数据时失败，因为没有授权写入权限
62 mysql> insert into student values (null, 'test', '500001', 18, 1, '2000-06-
63 ERROR 1142 (42000): INSERT command denied to user 'bit'@'localhost' for table
64 'student'
65

```

- 查看bit@localhost用户权限，USAGE 表示在指定的priv_level上没有权限

```

1 mysql> show grants for 'bit'@'localhost';
2 +-----+
3 | Grants for bit@localhost |
4 +-----+
5 | GRANT USAGE ON *.* TO `bit`@`localhost` |
6 | GRANT SELECT ON `java01`.* TO `bit`@`localhost` |
7 +-----+
8 2 rows in set (0.00 sec)

```

- 为bit@localhost用户授权于java01数据库的所有权限

```

1 # ALL为所有权限
2 mysql> grant ALL on java01.* to 'bit'@'localhost';
3 Query OK, 0 rows affected (0.01 sec)
4
5 # 查看用户权限
6 mysql> show grants for 'bit'@'localhost';
7 +-----+
8 | Grants for bit@localhost |
9 +-----+
10 | GRANT USAGE ON *.* TO `bit`@`localhost` |
11 | GRANT ALL PRIVILEGES ON `java01`.* TO `bit`@`localhost` |
12 +-----+
13 2 rows in set (0.00 sec)

```

```

14
15 # 插入一条新记录
16 mysql> insert into student values (null, 'test', '500001', 18, 1, '2000-06-
    08', 1);
17 ERROR 1142 (42000): INSERT command denied to user 'bit'@'localhost' for table
    'student'
18
19 # 不成功说明权限没有生效，需要刷新权限，如果还不成功可以重新登录
20 mysql> flush privileges;
21 Query OK, 0 rows affected (0.01 sec)
22
23 # 重新执行插入操作
24 mysql> insert into student values (null, 'test', '500001', 18, 1, '2000-06-
    08', 1);
25 Query OK, 1 row affected (0.02 sec)
26
27 # 查看记录
28 mysql> select * from student;
29 +-----+-----+-----+-----+-----+-----+-----+
30 | id | name   | sno   | age | gender | enroll_date | class_id |
31 +-----+-----+-----+-----+-----+-----+-----+
32 | 1 | 唐三藏 | 100001 | 18 | 1 | 1986-09-01 | 1 |
33 | 2 | 孙悟空 | 100002 | 18 | 1 | 1986-09-01 | 1 |
34 | 3 | 猪悟能 | 100003 | 18 | 1 | 1986-09-01 | 1 |
35 | 4 | 沙悟净 | 100004 | 18 | 1 | 1986-09-01 | 1 |
36 | 5 | 宋江   | 200001 | 18 | 1 | 2000-09-01 | 2 |
37 | 6 | 武松   | 200002 | 18 | 1 | 2000-09-01 | 2 |
38 | 7 | 李逵   | 200003 | 18 | 1 | 2000-09-01 | 2 |
39 | 8 | 不想毕业 | 200004 | 18 | 1 | 2000-09-01 | 2 |
40 | 12 | test   | 500001 | 18 | 1 | 2000-06-08 | 1 | # 新记录写入
    成功
41 +-----+-----+-----+-----+-----+-----+-----+
42 9 rows in set (0.00 sec)
43
44 # 删除操作
45 mysql> delete from student where id = 12;
46 Query OK, 1 row affected (0.01 sec)
47
48 # 删除成功
49 mysql> select * from student;
50 +-----+-----+-----+-----+-----+-----+-----+
51 | id | name   | sno   | age | gender | enroll_date | class_id |
52 +-----+-----+-----+-----+-----+-----+-----+
53 | 1 | 唐三藏 | 100001 | 18 | 1 | 1986-09-01 | 1 |
54 | 2 | 孙悟空 | 100002 | 18 | 1 | 1986-09-01 | 1 |
55 | 3 | 猪悟能 | 100003 | 18 | 1 | 1986-09-01 | 1 |
56 | 4 | 沙悟净 | 100004 | 18 | 1 | 1986-09-01 | 1 |

```

```

57 | 5 | 宋江      | 200001 | 18 | 1 | 2000-09-01 | 2 |
58 | 6 | 武松      | 200002 | 18 | 1 | 2000-09-01 | 2 |
59 | 7 | 李逵      | 200003 | 18 | 1 | 2000-09-01 | 2 |
60 | 8 | 不想毕业 | 200004 | 18 | 1 | 2000-09-01 | 2 |
61 +-----+-----+-----+-----+-----+-----+
62 8 rows in set (0.00 sec)

```

4.2 回收权限

4.2.1 语法

```

1 REVOKE [IF EXISTS] priv_type[, priv_type] ... ON priv_level
2   FROM 'user_name'@'host_name' [, 'user_name'@'host_name'] ...
3

```

4.2.2 示例

- 回收bit@localhost用户对于java01数据库的权限

```

1 mysql> REVOKE all on *.* from 'bit'@'localhost';
2 Query OK, 0 rows affected (0.01 sec)
3
4 # 刷新权限
5 mysql> flush privileges;
6 Query OK, 0 rows affected (0.01 sec)
7
8 # 查看权限，没有任何授权
9 mysql> show grants for 'bit'@'localhost';
10 +-----+
11 | Grants for bit@localhost |
12 +-----+
13 | GRANT USAGE ON *.* TO `bit`@`localhost` |
14 +-----+
15 1 row in set (0.00 sec)
16
17 # 查看所有数据库，没有可以访问的库
18 mysql> show databases;
19 +-----+
20 | Database |
21 +-----+
22 | information_schema |
23 | performance_schema |
24 +-----+

```

25 2 rows in set (0.00 sec)

已特戒业